

Notes: Xu (QJE, 2022)

Reshaping Global Trade: The Immediate and Long-Run Effects of Bank Failures

Contents

1	Big picture	3
2	Data and measurement (key objects)	3
2.1	British bank lending and failure exposure	3
2.2	Trade data	4
2.3	Units of observation and summary statistics	4
2.4	Important measurement choices	4
3	Models and empirical specifications (core equations + interpretation)	5
3.1	Naive finance-export relationship	5
3.2	Bartik-style exposure to failed banks	5
3.3	Testing whether failing banks sorted into different locations	5
3.4	Conceptual framework: sunk costs, substitutability, and finance	6
3.5	Immediate short-run effects on shipping	6
3.6	Entry and exit on the extensive margin	6
3.7	Long-run bilateral market-share specification	7
3.8	Structural-gravity interpretation	7
3.9	Indirect exposure and importers' re-sourcing	7
4	Identification and instruments (what makes the estimates credible)	7
4.1	Why the 1866 shock is useful	7
4.2	Narrative and quantitative evidence on bank failures	8
4.3	Formal identifying restriction	8
4.4	Evidence supporting the restriction	8
5	Tables and figures	9
5.1	Figure 1: Geography of banking and trade	9
5.2	Table 1: Summary statistics	9
5.3	Table 2: Pre-crisis comparison of bank characteristics	9

5.4	Table 3: Correlation between bank failures and pre-crisis location characteristics . .	10
5.5	Table 4: Intensive-margin effect on shipping	10
5.6	Table 5: Extensive-margin effect of exposure to bank failures	11
5.7	Figure 2: Aggregate exports for above- versus below-average exposure	11
5.8	Figure 3: Persistent effect of financing shock on exporter market share	13
5.9	Figure 4: Reduced form versus structural gravity	13
5.10	Figure 5: Decomposition of the effect of indirect exposure on importers	14
5.11	Figure 6: Persistent effects within groups of countries with similar exports	14
6	Short summary	15
7	Conclusion	15
7.1	Core conclusion (what the paper establishes)	15
7.2	Economic intuition	15
7.3	Takeaway	15

1 Big picture

- **Question.** Can a temporary banking crisis permanently reshape international trade patterns, or are trade effects of financial shocks only short-lived?
- **Setting.** The paper studies the 1866 London banking crisis, the first modern global banking crisis. Britain was then the center of the global financial system, and British multinational banks operated in countries accounting for about 98% of world exports.
- **Core empirical idea.** Locations differed in how dependent they were on London-headquartered British banks before the crisis. When some of those banks failed after the collapse of Overend & Gurney, locations more exposed to the failed banks experienced a sharper contraction in trade finance.
- **Immediate result.** A one-standard-deviation increase in exposure to failed banks lowers exports by about 8.5% one year later. More exposed exporters also have fewer trade partners and are less likely to begin exporting.
- **Long-run result.** The shock does not wash out. Importers buy significantly less from more exposed exporters for roughly four decades. In the baseline bilateral specification, a one-standard-deviation increase in exposure implies about 18% lower annual exports to a destination after the crisis.
- **Mechanism.** The paper emphasizes trade finance, sunk costs of establishing relationships, and substitutability across exporters. When incumbent suppliers become more expensive because finance tightens, importers form new relationships. Once these new links are in place, exposed exporters struggle to win back market share.
- **Why persistence is plausible.** Exporters may keep old relationships alive because re-entry is costly, but they become less able to enter *new* destinations. Meanwhile, importers expand along the extensive margin toward alternative suppliers.
- **Additional evidence.** Persistent losses are stronger for exporters selling more substitutable goods, for exporters with less access to alternative finance, and for more distant trade relationships. Imports are not significantly affected, which points to a supply-side trade-finance channel rather than a consumption channel.
- **Contribution.** The paper provides rare causal evidence that a temporary financial shock can create hysteresis in global trade patterns, not merely a short-run trade slump.

2 Data and measurement (key objects)

2.1 British bank lending and failure exposure

- **Core archival source.** The paper digitizes more than 11,000 handwritten Bank of England discount-window loan records from 1865–1866.
- **Bank sample.** These records cover 128 multinational banks headquartered in London. Of these, 21 failed during the crisis, so around 16% of the multinational banking sample ceased operations.

- **Geographic detail.** The records identify the bank originating the loan, the city where the loan was extended, loan amount, and discount date. This makes it possible to map pre-crisis British bank finance across cities around the world.
- **Exposure measure.** For each location, the paper computes pre-crisis bank shares and aggregates them using bank-failure indicators. This yields a continuous location-level measure of exposure to failed British banks.
- **Additional bank data.** The paper also collects bank balance sheets, shareholder-meeting transcripts, director and manager names, and narrative accounts to study why some banks failed and whether those failures were related to bank solvency or to local export fundamentals.

2.2 Trade data

- **Country-level panel.** The long-run analysis uses a country-level bilateral export panel from 1850–1914, assembled from historical trade statistics.
- **Industry composition.** For 1865, the paper codes countries’ exports into two-digit SITC industries, which helps measure substitutability across exporters and test heterogeneity by export bundle.
- **Port-level panel.** The short-run analysis uses daily shipping movements from the *Lloyd’s List* newspaper over a two-year window around the crisis.
- **Scale of shipping data.** The *Lloyd’s List* data contain more than 400,000 ship-movement observations. These are aggregated to country-level exports, port-level exports, and port-destination trade flows.

2.3 Units of observation and summary statistics

- **Ports.** The short-run shipping panel covers 289 ports. The average port has exposure of 0.07, about 119 outgoing ships in the pre-crisis period, around 7.6 destination countries, and about 6 British banks versus 0.45 non-British banks.
- **Countries.** The country sample covers 55 countries. The average country has exposure of 0.09, exports of about £12.2 million, around 3.9 destination countries, and roughly 13.5 British banks versus 1.26 non-British banks.
- **Interpretation.** British banks were globally pervasive, but exposure to failure varied substantially within and across countries. That cross-sectional variation is the empirical lever of the paper.

2.4 Important measurement choices

- **Port-city matching.** In the port-level analysis, each port is matched to the nearest city of finance by geodesic distance. Ports more than 500 km away from a finance city are treated as effectively unexposed controls.
- **Indirect exposure for importers.** To study re-sourcing, the paper computes an importer’s indirect exposure as the pre-crisis trade-share-weighted average exposure of its trade partners.

- **Why imports matter.** If the crisis mainly hit exporters' working capital, imports should not fall for countries simply because their *own* exposure rose. That is exactly what the paper finds.

3 Models and empirical specifications (core equations + interpretation)

3.1 Naive finance-export relationship

The object of interest is the causal effect of financing on exports at location l :

$$\ln(EX_l) = \alpha + \gamma \ln(Finance_l) + \Gamma' X_l + \varepsilon_l. \quad (1)$$

- **Interpretation.** EX_l is exports and $Finance_l$ is the amount of financing available to exporters.
- **Why OLS is not credible.** Finance is an equilibrium outcome, so weak places may both demand less finance and export less. Reverse causality is also possible if weak exporters damage banks' balance sheets.
- **Goal of the paper.** Replace endogenous local finance with plausibly exogenous exposure to London bank failures.

3.2 Bartik-style exposure to failed banks

Let $z_{lb,pre}$ denote bank b 's pre-crisis share in location l , and let $Failure_b$ indicate whether bank b failed in 1866. Exposure is

$$Fail_l = \sum_b z_{lb,pre} \mathbf{1}\{Failure_b\}. \quad (2)$$

The reduced-form export equation is

$$\ln(EX_{lt}) = \beta(Fail_l \times Post_t) + \Gamma' X_{lt} + \varepsilon_{lt}. \quad (3)$$

- **Interpretation of (2).** This is a shift-share/Bartik design. The shares are pre-crisis bank dependence; the shocks are bank failures in London.
- **Interpretation of (3).** β is the reduced-form semi-elasticity of exports with respect to exposure to failed banks after the crisis.
- **What varies.** Two locations can have the same overall British-bank presence but very different exposure if their banks had different failure rates.

3.3 Testing whether failing banks sorted into different locations

At the bank level, the paper regresses bank failure on the average characteristics of the locations where that bank operated:

$$\mathbf{1}\{Failure_b\} = \alpha + \beta \bar{X}_b + \varepsilon_b. \quad (4)$$

- **Purpose.** This is a balance test for the validity of the shift-share design.
- **Interpretation.** If banks that failed were systematically concentrated in places with weak export fundamentals, then the reduced-form estimates would confound finance with pre-existing location trends.
- **Result.** Most observable location characteristics do *not* predict failure exposure.

3.4 Conceptual framework: sunk costs, substitutability, and finance

The paper’s conceptual framework is verbal in the main text and formally developed in the Online Appendix. It is a Melitz-style environment in which external finance enters exporters’ marginal costs and relationship formation involves sunk costs.

- **Core assumptions.** Establishing a trade relationship is costly; exporters need working capital during shipment; many goods are highly substitutable across countries.
- **Finance channel.** A negative financing shock raises exporters’ prices and lowers profits.
- **Prediction 1.** Exposed exporters sell less in existing destinations because they become less competitive.
- **Prediction 2.** Exposed exporters are less likely to enter new destinations because higher financing costs lower expected profits net of sunk entry costs.
- **Prediction 3.** Importers facing higher prices from incumbent suppliers form new trade relationships.
- **Prediction 4.** Once new suppliers enter, exposed exporters can lose market share persistently, especially when goods are substitutable and switching is easy on the importer’s side.

3.5 Immediate short-run effects on shipping

At the country level, the short-run difference-in-differences specification is

$$\ln(S_{ot}) = \alpha Post_t + \beta(Fail_o \times Post_t) + \gamma_o + \Gamma'(X_o \times Post_t) + \varepsilon_{ot}. \quad (5)$$

At the port level, the specification is

$$\ln(S_{pot}) = \beta(Fail_{po} \times Post_t) + \Gamma'(X_{po} \times Post_t) + \alpha_p + \gamma_{ot} + \varepsilon_{pot}. \quad (6)$$

- **Dependent variable.** S is the number of ships departing in the pre- versus post-crisis period.
- **Why the port design matters.** γ_{ot} absorbs country-by-period shocks, so identification comes from comparing ports within the same country.
- **Interpretation of β .** Immediate semi-elasticity of export shipping to exposure to failed banks.

3.6 Entry and exit on the extensive margin

To study entry into and exit from exporting, the paper estimates

$$\Pr(E_{po} = 1) = \alpha + \beta Fail_{po} + \gamma_o + \Gamma' X_{po} + \varepsilon_{po}. \quad (7)$$

- **Interpretation.** E_{po} is either an entry indicator or an exit indicator for port p .
- **Prediction.** If sunk costs matter, the shock should reduce entry more than it raises exit.
- **Result.** That is exactly what the paper finds.

3.7 Long-run bilateral market-share specification

The core long-run specification is

$$\ln(EX_{odt}) = \beta_t Fail_o + \Gamma' X_{ot} + \gamma_o + \gamma_{dt} + \theta_t \ln(dist_{od}) + \varepsilon_{odt}. \quad (8)$$

- **Dependent variable.** Bilateral exports from origin o to destination d at time t .
- **Why importer-year fixed effects matter.** γ_{dt} absorb demand shocks in each destination-year, so β_t is a within-destination relative market-share effect across exporters.
- **Dynamic interpretation.** The annual or five-year coefficients β_t trace pretrends, immediate impact, and persistence.
- **Distance control.** Time-varying distance effects absorb changing sensitivity of trade to geography over time.

3.8 Structural-gravity interpretation

The paper also estimates a structural-gravity variant of equation (8) by approximating multilateral resistance terms following Baier and Bergstrand (2009).

- **Purpose.** Translate the reduced-form export response into a trade-cost elasticity under the stronger assumption that the banking shock acts purely through trade costs.
- **Result.** On the common sample, the structural estimate is larger in magnitude than the reduced form (-1.88 versus -1.50), consistent with general-equilibrium amplification, though not statistically different from it.
- **Author's stance.** Xu treats the reduced form as the baseline because the structural interpretation requires stronger assumptions.

3.9 Indirect exposure and importers' re-sourcing

To study how importers react to suppliers' financing shocks, the paper estimates

$$\ln(IM_{dt}) = \beta_t Fail_{d,pre} + \gamma_d + \gamma_t + \varepsilon_{dt}. \quad (9)$$

- **Indirect exposure.** $Fail_{d,pre}$ is the pre-crisis trade-share-weighted average exposure of importer d 's trade partners.
- **Interpretation.** This captures whether importers whose incumbent suppliers were hit harder later expanded along the extensive margin and shifted market shares toward new suppliers.
- **Core finding.** Total imports do not move much, but the number of source countries rises, imports per source fall, and the share coming from old relationships declines.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the 1866 shock is useful

- **Crisis origin.** The shock began with the unexpected failure of Overend & Gurney, a prestigious London discount house whose collapse was tied to fraudulent mismanagement rather than a

global trade downturn.

- **Panic environment.** The failure triggered severe runs on London’s deposit-issuing banks and a flight to liquidity. The crisis therefore looks like a panic rather than an orderly re-pricing of obviously weak banks.
- **Why that matters.** If failures were caused by local real shocks in the banks’ operating regions, the trade results would be much harder to interpret causally.

4.2 Narrative and quantitative evidence on bank failures

- **Main predictor of failure.** The strongest observable predictor is a public connection to Overend & Gurney through bank managers or directors appearing on Overend’s shareholder list.
- **What does *not* predict failure.** Bank capital, deposits, leverage, reserve ratio, liquidity ratio, total size, and geographic specialization are broadly balanced between failed and non-failed banks.
- **Interpretation.** Depositors appear to have run from banks associated with Overend, not from banks that were observably more insolvent.

4.3 Formal identifying restriction

- **Key requirement.** The design does *not* require pure randomness of bank failures. It requires that failing banks were not systematically located in places whose export path would have worsened for other reasons.
- **Shock-share logic.** What must be orthogonal to export fundamentals is the bank-level failure shock, weighted by pre-crisis location shares.
- **Difference-in-differences interpretation.** Conditional on controls and fixed effects, there should be no simultaneous location-specific shocks that both reduce exports and make the banks operating there fail.

4.4 Evidence supporting the restriction

- **Balance tests.** Bank failures are mostly uncorrelated with pre-crisis port and country characteristics. At the port level, only higher UK export share and capital-city status show significant relationships; the paper controls for these directly.
- **No pretrends.** Countries more and less exposed to failed banks display parallel export trends before 1866.
- **Within-country design.** Port-level estimates with country-by-period fixed effects yield very similar coefficients to country-level estimates.
- **Additional robustness.** Results survive exclusion of cotton exporters, Poisson count specifications, controls for industry mix, military conflicts, exchange-rate regimes, financial crises, Suez effects, and placebo/randomization tests.

5 Tables and figures

5.1 Figure 1: Geography of banking and trade

Read:

- British bank lending before the crisis is globally dispersed, not concentrated in a few colonies or ports.
- Exposure to failed banks varies both across countries and within countries, which is what makes the port-level design possible.
- The map of shipping activity confirms that the trade data also have broad geographic coverage, so the paper is not about a narrow regional episode.

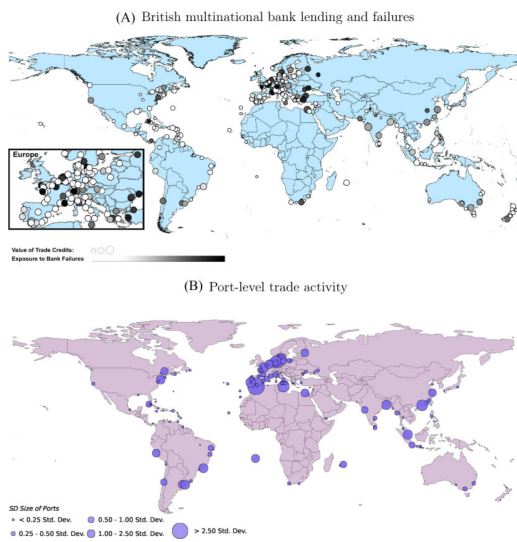


FIGURE I
Geography of Banking and Trade

TABLE I
SUMMARY STATISTICS: PORTS AND COUNTRIES

	Ports			Countries		
	Mean	Median	Std. dev.	Mean	Median	Std. dev.
Exposure to failed British banks	0.07	0.00	(0.19)	0.09	0.03	(0.14)
Exports	118.76	27.00	(215.53)	12.20	2.14	(31.89)
Share exports to United Kingdom	0.39	0.30	(0.34)	0.60	0.69	(0.38)
Destinations (no. countries)	7.63	5.00	(7.21)	3.85	2.00	(8.06)
Distance to destination (km k)	5.06	4.80	(3.31)	6.52	6.31	(3.51)
British banks	6.03	3.00	(7.54)	13.45	9.00	(13.82)
Non-British banks	0.45	0.00	(0.82)	1.26	0.00	(3.25)
Fraction in British Empire	0.34	0.00	(0.47)	0.33	0.00	(0.47)
N	289			55		

Notes: The table shows summary statistics from the port-level panel of shipping activity and the country-level panel of values of exports. All variables are measured at the end of 1865, before the crisis. "Exports" is measured by the number of ships departing for ports, and by the value of exports in millions of pounds sterling for countries. Share of exports to the United Kingdom is similarly calculated using the number of ships and values of exports.

5.2 Table 1: Summary statistics

Read:

- Exposure is economically meaningful but far from universal: mean exposure is 0.07 for ports and 0.09 for countries.
- British banks dominate foreign banking connections almost everywhere: ports average 6.03 British versus 0.45 non-British banks; countries average 13.45 British versus 1.26 non-British banks.
- Countries are very heterogeneous in export size and geographic reach, which is why the paper relies on fixed effects and weighted estimators rather than raw comparisons.

5.3 Table 2: Pre-crisis comparison of bank characteristics

Read:

TABLE II
PRECRISIS COMPARISON OF BANK CHARACTERISTICS

	All	Not failed	Failed	Diff
Panel A: Balance sheet characteristics (joint-stock banks)				
OG connection	0.16 (0.37)	0.11 (0.32)	0.37 (0.50)	-0.26 (0.09)***
Capital, authorized (£m)	1.42 (1.06)	1.39 (1.05)	1.56 (1.13)	-0.16 (0.28)
Capital, paid up (£m)	0.58 (0.38)	0.60 (0.37)	0.47 (0.41)	0.13 (0.10)
Deposits (£m)	2.22 (2.73)	2.29 (2.82)	1.85 (2.37)	0.44 (1.14)
Reserve fund (£m)	0.13 (0.12)	0.13 (0.12)	0.13 (0.16)	-0.01 (0.04)
Total size (£m)	4.76 (6.08)	4.96 (6.38)	3.83 (4.65)	1.13 (1.88)
Leverage ratio	0.06 (0.05)	0.06 (0.05)	0.06 (0.06)	0.00 (0.02)
Reserve ratio	0.03 (0.02)	0.03 (0.02)	0.03 (0.02)	-0.00 (0.01)
Liquidity ratio	0.14 (0.11)	0.14 (0.09)	0.12 (0.15)	0.02 (0.03)
N	100	81	19	100
Panel B: Other characteristics (all banks)				
OG connection	0.16 (0.37)	0.13 (0.34)	0.33 (0.48)	-0.20 (0.09)**
Trade finance (£k)	106.65 (246.58)	112.21 (263.75)	72.23 (126.50)	39.98 (58.98)
Age (years)	36.95 (53.22)	41.31 (56.48)	14.76 (21.34)	26.55 (12.53)**
Cities (#)	13.91 (23.11)	15.01 (24.74)	8.29 (10.34)	6.72 (5.51)
Countries (#)	6.98 (7.78)	7.16 (8.00)	6.05 (6.60)	1.11 (1.86)

TABLE II
CONTINUED

	All	Not failed	Failed	Diff
Regions (#)	2.58 (1.58)	2.62 (1.61)	2.38 (1.43)	0.24 (0.38)
Asia (£k)	44.31 (169.41)	47.56 (183.47)	27.79 (59.26)	19.76 (40.66)
Africa (£k)	7.83 (23.60)	7.14 (21.85)	11.39 (31.47)	-4.26 (5.64)
N. America (£k)	10.89 (37.47)	12.30 (40.47)	3.73 (13.05)	8.56 (8.95)
S. America (£k)	7.44 (34.26)	8.38 (37.22)	2.65 (9.15)	5.73 (8.19)
Oceania (£k)	6.39 (17.24)	7.06 (18.51)	2.98 (7.54)	4.07 (4.12)
Europe (£k)	25.98 (61.57)	26.48 (64.95)	23.40 (41.28)	3.08 (14.75)
British Emp (£k)	60.39 (159.09)	65.47 (171.52)	34.50 (62.90)	30.96 (38.02)
UK (£k)	12.11 (39.69)	14.26 (43.10)	1.18 (2.73)	13.07 (9.44)
N	128	107	21	128

Notes: The table shows bank-level balance sheet characteristics for banks that failed and did not fail. All variables are measured at the end of 1865 before the crisis. Balance sheet variables were only published for publicly traded banks; these are reported separately in Panel A. "Not failed" and "Failed" refers to whether a bank suspended or closed during the crisis. Means are reported first, and standard deviations are given in parentheses. "Diff" refers to the difference in means between groups. Standard errors are reported in parentheses for the "Diff" column. £k denotes units of thousands of pounds sterling. £m denotes units of millions of pounds sterling. Leverage ratio is defined as capital (paid and reserves) divided by total assets. Reserve ratio is defined as reserve assets divided by deposit liabilities. Liquidity ratio is defined as cash, gold, and short-term bills divided by total assets. "Cities," "countries," and "regions" count the number of unique geographic locations banks operate in. Significance is marked by * $p < .1$, ** $p < .05$, *** $p < .01$. Sources: Bank of England Archives CSM, Banker's Magazine, The Economist.

- Failed and surviving banks look similar on balance-sheet measures such as capital, deposits, size, leverage, liquidity, and reserves.
- The strongest difference is the Overend-connection variable: banks connected to Overend are much more likely to fail.
- Failed banks are also younger on average, but the evidence does not indicate systematic concentration in weaker export regions.
- This table supports the panic interpretation: failure was not simply the mechanical outcome of weak observable fundamentals.

5.4 Table 3: Correlation between bank failures and pre-crisis location characteristics

TABLE III
CORRELATION BETWEEN BANK FAILURES AND PRECRISIS LOCATION CHARACTERISTICS

	Ships (1)	Share to UK (2)	Distance to London (3)	Brit empire (4)	Foreign banks (5)	Destinations (6)	Capital city (7)
Panel A: Port characteristics							
	0.0273 (0.0358)	0.171*** (0.0335)	-0.0641 (0.0394)	-0.0723 (0.0411)	-0.0341 (0.0262)	-0.0612 (0.0337)	-0.131*** (0.0384)
N	122	122	122	122	122	122	122
Exports val. (1)		Δ Exports (2)	Share to UK (3)	Gold (4)	Silver (5)	Binetallic (6)	Conflict: any (7)
Panel B: Country characteristics							
	0.129 (0.0677)	-0.0269 (0.0246)	-0.163 (0.0346)	-0.0387 (0.0353)	-0.00457 (0.0449)	0.0434 (0.0432)	0.00720 (0.0456)
N	128	128	128	128	128	128	128

TABLE III
CONTINUED

	Cotton, raw (1)	Cotton, manu. (2)	Cotton (3)	Grains (4)	Bullion (5)	Coffee (6)	Alcohol (7)	Commodities share (8)
Panel C: Country characteristics: exports composition								
	0.0428 (0.0299)	-0.0563 (0.0357)	-0.0438 (0.0339)	0.0571 (0.0301)	-0.0624 (0.0344)	-0.0166 (0.0252)	-0.0446 (0.0360)	-0.0253 (0.0302)
N	128	128	128	128	128	128	128	128

Notes: The table reports estimates from the bank-level regression of bank exposure to location characteristic precrisis on bank failure rates. The dependent variable is $I(\text{Failure}_i)$, the measure of bank failure. The independent variable of interest X_i is the share-weighted exposure of banks to location characteristic, normalized to have zero mean and unit variance. Each column is labeled with the independent variable. The coefficients are interpreted as the increase in the probability that a bank fails given a standard deviation increase in the average bank exposure to a particular characteristic. Panel A includes location characteristics from the port level. Panels B and C include country-level characteristics like the monetary standard and presence of conflict in the exporting country in 1865/1866, and the industry composition of exports in 1865. The effective sample size is the inverse of the HH of shock-level average exposure: $\frac{1}{\frac{1}{N} + \frac{1}{N_{\text{port}}}}$ and is equal to 27.55 for ports and 28.05 for countries. Regressions are weighted by the average location's exposure to bank b . * $p < .1$, ** $p < .05$, *** $p < .01$. As discussed in Borovik, Hall, and Jarrell (2022), another advantage of transforming the balance tests into shock-level (bank level) regressions is that it makes it clear which shocks (banks) are the most relevant for the results. In Panel A, there are 122 observations instead of the full 128 because on banks operated in cities that were not the closest city for any port, so they do not contribute to the port-level exposure measures.

Read:

- Most pre-crisis location characteristics do not predict whether the banks operating there fail.
- At the port level, banks serving ports more tied to the UK are slightly more likely to fail, while those serving capital-city ports are less likely to fail; these are controlled for later.
- At the country level, export values, export growth, conflict exposure, monetary standard, and export composition are broadly unrelated to bank failures.
- Economically, this table says the treatment is not obviously picking up pre-existing trade weakness.

5.5 Table 4: Intensive-margin effect on shipping

Read:

- The country-level short-run coefficient is about -0.606 , and the port-level coefficient with the full control set is about -0.656 .

- These magnitudes imply that a one-standard-deviation increase in exposure lowers exports by about 8.5% after one year, with very similar estimates across country and within-country port designs.
- The result survives excluding cotton exporters, using count-data methods, and aligning the post period with news-arrival dates.
- The similarity across columns strengthens the claim that omitted country-level shocks are not driving the result.

TABLE IV

INTENSIVE-MARGIN EFFECT OF BANK FAILURES ON SHIPPING

$$\text{Col 1: } \ln(S_{it}) = \alpha \text{Post}_t + \beta \text{Fail}_{it} \times \text{Post}_t + \Gamma' X_{it} \times \text{Post}_t + \gamma_0 + \gamma_{it} + \epsilon_{it}$$

$$\text{Col 2-7: } \ln(S_{it}) = \beta \text{Fail}_{it} \times \text{Post}_t + \Gamma' X_{it} \times \text{Post}_t + \alpha_0 + \alpha_{it} + \gamma_{it} + \epsilon_{it}$$

	Country	Port		Excl. cotton	Poisson	News dates	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Fail _{it} × post	-0.606*** [0.214]						
Fail _{it} × post		-0.727*** [0.251]	-0.724*** [0.159]	-0.656*** [0.210]	-0.554*** [0.204]	-0.966*** [0.169]	-0.592*** [0.208]
Capital city × post				Y	Y	Y	Y
Age of banks × post				Y	Y	Y	Y
OG link × post				Y	Y	Y	Y
Share to UK × post				Y	Y	Y	Y
Country _o FE	Y						
Port _o FE		Y	Y	Y	Y	Y	Y
Country _o × post FE			Y	Y	Y	Y	Y
N	108	578	578	578	494	578	570
Ports		289	289	289	247	289	289
Clusters	54	54	54	54	50	54	54

Notes: The table reports estimates from the difference-in-difference regressions from the two-period panel of port-level shipping activity in the year before and after the crisis. The dependent variable in columns (1)–(5) and (7) is the log of the total number of ships departing in each period. In column (6), a Poisson estimator is used as the dependent variable is the count of ships. Fail_{it} is the share of the port's banks that failed during the crisis. The mean of Fail_{it} is 0.07, and the standard deviation is 0.2. Post is a dummy for the postcrisis year that takes the value of 1 after May 1866 and 0 otherwise. The time-invariant control variables are measured in 1865 and interacted with the post dummy. They include an indicator for the port being a capital city in the country, the average age of banks, the fraction of banks with an Overseas connection, and the share of shipping in the United Kingdom in the preshock period. The sample is restricted to ports active in both the pre- and postperiod. In column (5) the sample excludes major cotton exporting countries (defined as Brazil, Egypt, India, and the United States). In column (7), the "post" crisis period varies at the port level based on the news arrival dates of the crisis from London. Standard errors in brackets are clustered by country of origin. * $p < .1$, ** $p < .05$, *** $p < .01$.

TABLE V
EXTENSIVE-MARGIN EFFECT OF EXPOSURE TO BANK FAILURES

	Country: dest (1)	Country: avg ships (2)	Ports: dest (3)	I(Port entry) (4)	I(Port exit) (5)
Fail _{it} × post	-0.592*** [0.196]	-0.0138 [0.299]	-0.409*** [0.149]		
Fail _{it} × post				-0.303*** [0.0846]	-0.00227 [0.169]
Port controls × post			Y		
Port controls				Y	Y
Port _o FE			Y		
Country _o × post FE			Y		
Country _o FE	Y	Y		Y	Y
N	108	108	574	331	318
Ports			289	331	318
Clusters	54	54	54	55	54

Notes: The table reports estimates of the effect of the exposure to bank failures on the extensive margin of shipping activity. The dependent variable in columns (1) and (3) is the log number of unique destinations accessed by countries and ports, respectively. The dependent variable in column (2) is the log of the average number of ships per destination from a country. The sample in columns (1)–(3) is restricted to ports that were active in both the preshock and the posts shock periods. The dependent variable in column (4), I(Port entry), is a binary variable that takes the value of 1 for a port that was not active in the preshock period and became active in the posts shock period, and 0 otherwise. The dependent variable in column (5), I(Port exit), is a binary variable for a port that was active in the preshock period and became inactive in the posts shock period. The sample sizes in columns (4) and (5) reflect the number of ports that were active in the postperiod (for Entry) or preperiod (for Exit). All variables are defined the same way as in Table IV. Standard errors in brackets are clustered by country of origin. * $p < .1$, ** $p < .05$, *** $p < .01$.

5.6 Table 5: Extensive-margin effect of exposure to bank failures

Read:

- More exposed countries and ports end up with fewer destinations: the coefficients are about -0.592 at the country level and -0.409 at the port level.
- Average ships per destination do not fall significantly, which means the total-export decline is mostly about fewer relationships rather than much smaller flows within each surviving route.
- Entry into exporting falls sharply (coefficient around -0.303), but exit does not rise meaningfully.
- This asymmetry is exactly what one expects if sunk entry costs matter.

5.7 Figure 2: Aggregate exports for above- versus below-average exposure

Read:

- Before 1866, the two groups have very similar export growth paths, which is visual evidence against differential pretrends.
- After 1866, aggregate exports diverge immediately and never reconverge.
- The figure shows the big-picture level effect: the crisis is associated not just with a temporary dip, but with a permanently lower export path for exposed countries.

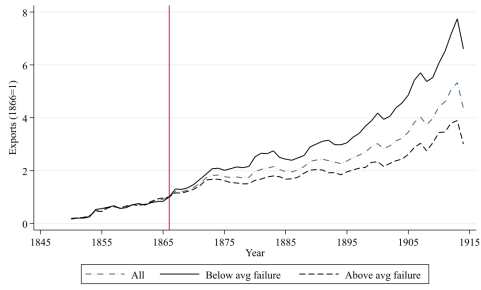


FIGURE II

Aggregate Exports, Grouping Countries by Above- and Below-Average Exposure to Bank Failures

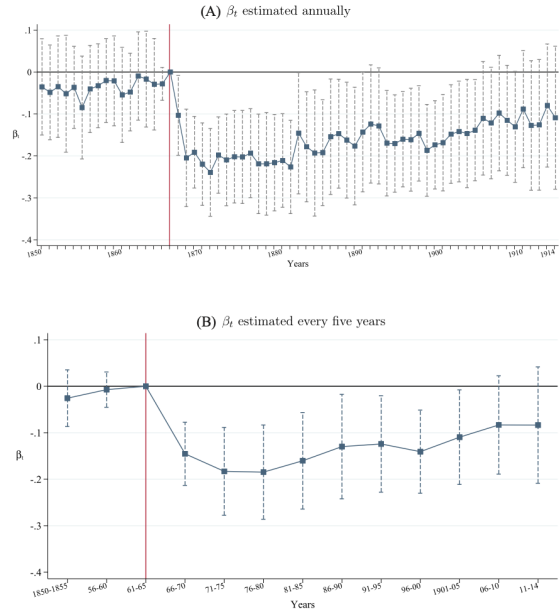


FIGURE III

Persistent Effect of Financing Shock on Exporter Market Share

$$\ln(EX_{odt}) = \beta_t \text{Fail}_o + \Gamma' X_{ot} + \gamma_o + \gamma_{dt} + \theta_t \ln(\text{dist})_{od} + \varepsilon_{odt}$$

Figure plots the β_t point estimates and 95% confidence intervals for the specification given in [equation \(8\)](#) estimated on the country-level panel of trade. β_t is the treatment coefficient on the effect of exposure to failed banks on exports in each group of years. Point estimates and standard errors are scaled by the mean of treatment, so the magnitudes should be interpreted as the effect for the average exporter. The dependent variable is the log value of exports. The specification includes origin country o FE, destination country-year dt FE, and time-varying controls for the bilateral distance between countries. Standard errors are clustered by the origin country. See [Online Appendix Table G12](#), column (1) for the point estimates.

5.8 Figure 3: Persistent effect of financing shock on exporter market share

Read:

- The event-study coefficients are close to zero before the crisis and strongly negative after it.
- The post-crisis market-share losses last for about four decades; in the two-period version, the unscaled post coefficient is about -1.32 .
- Scaled to average exposure, the paper interprets this as roughly 12% lower annual exports to a destination after the crisis.
- This is the central persistence result of the paper.

5.9 Figure 4: Reduced form versus structural gravity

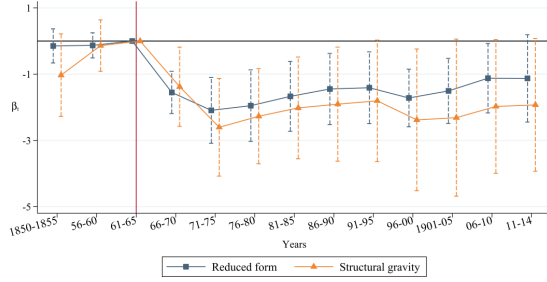


FIGURE IV

Persistent Effects: Comparing Baseline with Structural Version

Figure plots the point estimates and 95% confidence intervals for two separately estimated regressions. The one labeled “Reduced form” follows the specification in equation (8) that estimates the reduced-form β without controlling for multilateral resistance terms. The one labeled “Structural gravity” corresponds to the coefficients in Online Appendix Table F1, column (4) that estimates both the ϕ and ψ in the estimating equation (F7) in the Online Appendix. Standard errors are clustered by the origin country.

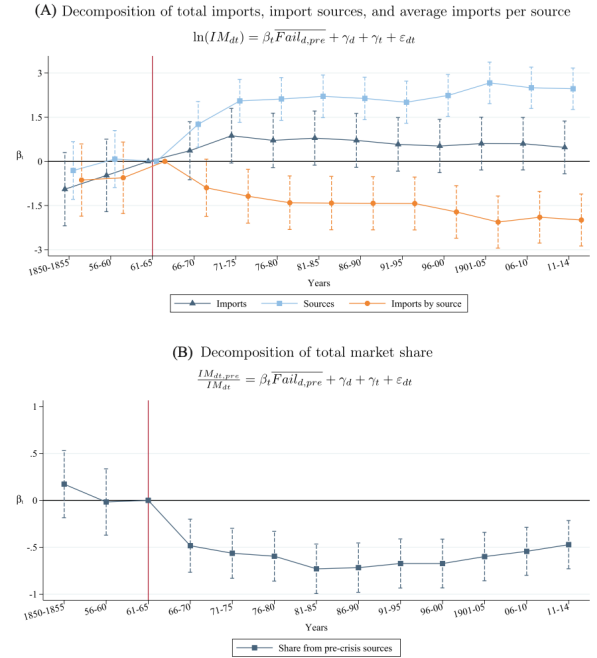


FIGURE V

Decomposition of Effect of Indirect Exposure on Importers

Panel A plots the β_t point estimates and 95% confidence intervals for three separately estimated regressions of the specification written above and the dependent variable indicated. The specification includes destination (importer) d and year fixed effects. The dependent variable is the log of total imports (dark blue triangle), the log of the number of trade partners (light blue square), and the log of average imports per source (orange circle). $\overline{Fail}_{d,pre} = \sum_o s_{od} Fail_o$, where $s_{od} = \frac{IM_{do}}{IM_d}$. The estimated effects are not scaled by average indirect exposure, so they should be interpreted as the effect if all of an importer's trade partners precrisis were fully exposed to the bank failures. Panel B has the share of total imports from precrisis trade partners as the dependent variable. The observations are weighted by country size proxied by the number of trading partners to most closely mirror the aggregates in Figure II.

Read:

- The structural-gravity estimate is more negative than the reduced form on the common sample, consistent with general-equilibrium amplification of trade-cost shocks.
- But the two sets of estimates are qualitatively very similar, and the structural estimate is not statistically different from the reduced-form estimate.

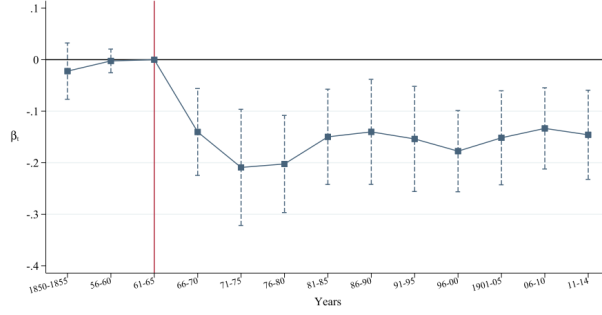


FIGURE VI

Persistent Effects within Groups of Countries with Similar Exports
 $\ln(EX_{odt}) = \beta_t \text{Fail}_o + \Gamma' X_{ot} + \gamma_o + \gamma_{dt} + \psi \text{Region}_{ot} + \theta_t \ln(\text{dist})_{od} + \varepsilon_{odt}$

Figure plots the point estimates and 95% confidence intervals for the specification given above estimated on the country-level panel of trade. β_t is the treatment coefficient on the effect of exposure to failed banks on exports in each group of years. Point estimates and standard errors are scaled by the mean of treatment, so the magnitudes should be interpreted as the effect for the average exporter. The dependent variable is the log value of exports. The specification includes origin-country region-year FE, origin country o FE, destination country-year dt FE, and time-varying controls for the bilateral distance between countries. Standard errors are clustered by the origin country. See [Online Appendix Table G12](#), column (8) for the point estimates.

- This figure is useful because it shows that the persistence result does not depend on one specific interpretation of the coefficient.

5.10 Figure 5: Decomposition of the effect of indirect exposure on importers

Read:

- Importers indirectly exposed through their suppliers do not significantly reduce total imports.
- Instead, they increase the number of source countries and reduce imports per source.
- The share of imports coming from pre-crisis suppliers falls persistently, which mirrors the exporters' market-share losses.
- This is the clearest direct evidence for the re-sourcing mechanism.

5.11 Figure 6: Persistent effects within groups of countries with similar exports

Read:

- When exporters are compared only with other exporters from the same region, the losses remain highly persistent.
- This strengthens the interpretation that importers substitute toward similar producers when exposed suppliers become more expensive.
- It also weakens the idea that the results merely reflect broad regional trends unrelated to product substitutability.

6 Short summary

- **Conclusion.** The 1866 London banking crisis caused an immediate export contraction in exposed places and a persistent reallocation of global trade toward less exposed exporters.
- **Mechanism.** Trade finance mattered because it affected exporters' marginal costs, and the resulting relative price changes induced importers to build new trade relationships that exposed exporters could not easily reverse.
- **Empirical lesson.** A historical shift-share design can recover not just short-run treatment effects but decades-long hysteresis when the shock is plausibly exogenous and the panel is long enough.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Financial crises can reshape trade patterns for decades, not only during the crisis itself.
- **Immediate magnitude.** A one-standard-deviation increase in exposure to failed British banks reduces exports by about 8.5% in the short run.
- **Long-run magnitude.** Importers buy about 18% less each year from exporters with a one-standard-deviation higher exposure, and these relative losses last roughly four decades.
- **Credibility.** The crisis looks like a panic centered on Overend & Gurney rather than a real shock hitting export regions. Balance tests, within-country estimates, and placebo exercises all support this interpretation.
- **Mechanism evidence.** Importers do not cut total imports; they increase the number of suppliers and reduce the share bought from old exposed partners. This is exactly the pattern implied by re-sourcing plus sunk costs.
- **Heterogeneity.** Losses are stronger when exporters are easier to substitute away from, when trade relationships are more distant, and when exporters lack alternative finance.

7.2 Economic intuition

- Trade finance is a working-capital input for exporters. When banks fail, exporters effectively face a higher marginal cost of reaching foreign buyers.
- Sunk costs matter because exporters can lose the race for *new* relationships even if they try to preserve old ones.
- Once importers switch toward alternative suppliers, the new network structure itself becomes persistent, generating hysteresis in market shares.

7.3 Takeaway

The paper's broader message is that history can matter in international trade through finance. A temporary financial disruption in the world's dominant banking center changed who traded with

whom for decades because it altered the timing of relationship formation during a period of global integration. In that sense, the crisis was not just a temporary liquidity event; it was a shock to the evolution of the world trading network.